

CASE STUDY

Raktdaan — Blood Donation Center

raktdaan.online • Full Stack MERN Web Application

PROJECT	Raktdaan Blood Donation Platform
WEBSITE	https://www.raktdaan.online
TYPE	Full Stack Web Application
ROLE	Full Stack Developer (Solo)
TECH STACK	React.js, Node.js, Express.js, MongoDB, TypeScript, Git

■ The Problem

Blood donation management in India is largely manual and highly fragmented. Patients in urgent need struggle to find matching donors quickly, and blood banks have no centralized digital system to manage requests efficiently. Families often rely on phone calls and word-of-mouth in life-or-death situations — wasting precious time.

■ The Solution

Built **Raktdaan** — a full-stack web application that digitizes and streamlines the entire blood donation process. The platform connects donors with patients/hospitals in real time, making it faster and easier to find life-saving blood.

■ Key Features Developed

■ Donor Registration	Users can register and manage their donor profile with blood group and location.
■ Smart Donor Search	Patients/hospitals can filter donors by blood group and city instantly.
■ Real-Time Requests	Blood requests are posted and matched to available donors in real time.
■ Admin Dashboard	Admins can manage all donor records, requests, and approvals centrally.
■ Responsive Design	Fully mobile-friendly — works seamlessly on all screen sizes.
■ Secure & Scalable	Built with secure APIs and a scalable architecture for future growth.

■ Tech Stack

Layer	Technology
Frontend	React.js + TypeScript
Backend	Node.js + Express.js
Database	MongoDB
Version Control	Git / GitHub
Deployment	Web Hosting / Cloud

■ Results & Impact

- Created a free, accessible platform to help save lives across India

- Significantly reduced time to find a matching blood donor
- Fully responsive — accessible from any mobile device
- Scalable architecture ready to expand to multiple cities and states
- Successfully deployed and live at raktdaan.online

■ Use This in Your Proposal

"I recently built Raktdaan (raktdaan.online), a full-stack blood donation platform using React.js, Node.js, Express.js and MongoDB. I handled everything from database design to frontend UI and live deployment. I can bring the same dedication and technical skill to your project."

Prepared by: Full Stack Developer • raktdaan.online • Pune, India